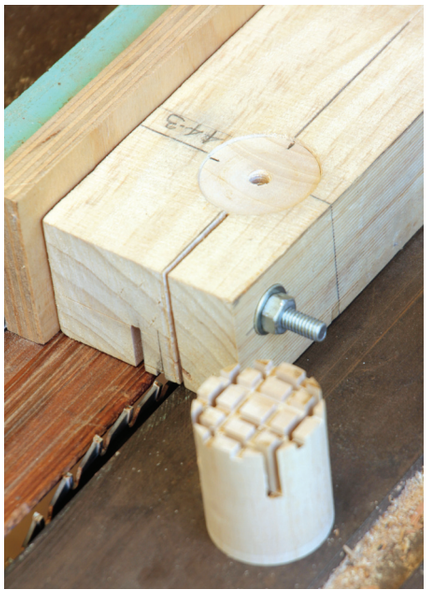
Making my design



The jig for cutting the rook crenels. The jig's vertical through hole is 1¾ inches diameter. A rook workpiece is clamped in the hole using the screw. The reference pencil marks allow the workpiece to be rotated in 90° increments

Sawing the slot in a bishop using a sloping jig that locks into the lathe banjo

WORKPIECE CHUCK

The third article in this series showed a homemade ½in diameter screwchuck. Screwchucks are excellent for faceplate work, but a workpiece that is small in diameter and smooth can be difficult to unscrew off. You can remove a stubborn workpiece by gripping it with a piece of fine-grained abrasive paper. One alternative is to instead use a chuck that has been neglected in recent years, the loose pin chuck. Manufactured versions are typically around 25mm in diameter, but this is obviously too large for most chessmen. My 'fixed' pin is here a length of ½in diameter steel bar held in a scroll chuck. The loose pin is just over 2mm diameter, and is cut from a wire coat hanger – this is handy because you can easily cut more to replace any loose pins, that get lost in the shavings. You can of course make your own loose pin chuck to any suitable diameter providing that you can source a drill to bore a hole of the same diameter in your chessman workpieces.

Although my pieces' forms – except for the pawn's – have enough volume low down to allow leading, I decided against it. I also decided to mount the men's workpieces on a commercial screw chuck. This allows simple rechucking for polishing. I therefore used a turning procedure similar to that described in the third article:

1. Rough each man's workpiece to a cylinder about 6mm oversize in both length and diameter. However, the rook workpieces must be turned to the rooks' finished length and accurately to the slot-sawing jig's hole diameter – in my case 44.5mm
2. Chuck each workpiece axially by the top end
3. Bore the hole for screw chucking – ¼in diameter for many commercial screw chucks
4. The piece which presents the greatest challenge is the rook because its base diameter exceeds the breadth of its battlemented top. The rooks therefore can't be turned from plain, dressed square sections whose thickness equals their battlement's breadth. There is also the challenge of carving the embrasures, also called crenels or crenelles, and producing the square recess in the top. To cut the crenels I clamped the workpiece top down into a sawing jig, and sawed a slot using a table saw. Before cutting each of the other three slots, I rotated the workpiece 90°
5. The slot in the bishops is substantially narrower than I could cut with my circular saw blade; it's also canted. To cut the slot I used a hand saw sharpened with rip teeth supported on and guided by the sloping-surface jig shown above
6. With each workpiece chucked on the commercial screw chuck, finish-turn and sand each man working from right to left. Turn away the central square upstanding section in the top of each rook to create the square recess
7. Carve each bishop's helmet's peak
8. Polish in the lathe as described in the first article
9. The hole in the bottoms of the men could be left unfilled, or be filled for example by using a mortar of lead shot and gap-filling adhesive as described in the previous article. The men's bottoms could also be covered with discs of thin leather or baize as described in article two

Conclusion

Although strongly based on the Staunton design, this set design illustrates that the scope to create new chess set designs is infinite even if all the pieces are required to be turned. ●

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